All American Eats

Deliverable 2: Requirement Specifications

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Nightfall Labs

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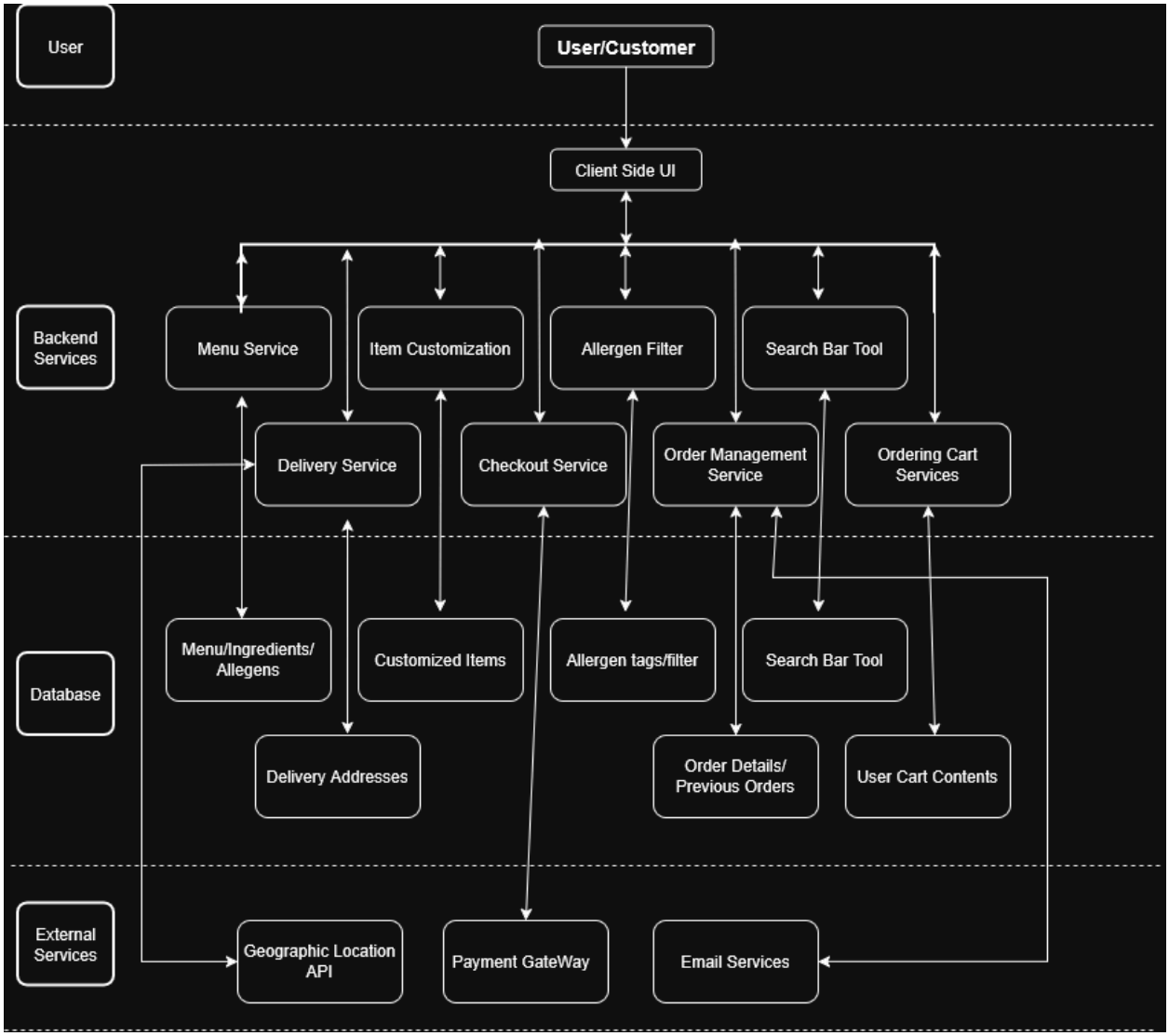
1. The overall structure of the system. This should contain a diagram and brief descriptions about the system structure including individual components (subsystems).

Diagram:



Overall System Architecture

All American Eats follows a layered client server architecture. The system consists of a client side user interface that communicates with the backend services through RESTful APIs. The backend services handle the business logic such as menu retrieval, filtering, customization, cart management, checkout processing, and order management. Data is stored in a centralized database, while the external services are used for processing payment, verifying location, and email notifications.

The system is divided into four primary layers:

1. **Presentation Layer (Client Side UI)** – Handles user interaction and display.
2. **Application Layer (Backend Services)** – Contains business logic and service coordination.
3. **Data Layer (Database)** – Stores menu, user, and order information.
4. **External Integration Layer** – Connects to third-party APIs such as payment gateways and location services.

This structure ensures scalability, maintainability, and separation of concerns.

Subsystems:

Client Side UI

- Display for menus and categories
- Customizable dishes
- Allergen and dietary restriction filter controls
- keyword search for navigation
- Item cart interactions
- Information gathering such as delivery address and contact information.

Menu Service

- Displays menu items from the database
- Shows accurate dish descriptions with their price.
- Have and out of stock option to remove items from menu if needed

- Display ingredient and allergy data with dishes

Item Customization

- Stores client customization options for each dish
- Adds additional costs to customization options
- Attach customization options to the clients cart

Allergen/Dietary Restrictions filter

- Working filter that remove any items containing the allergens that the client selects
- Flags items if they “may contain” an allergen
- Removes items from menu that contain an allergen

Search Bar Service

- Search bar accessible in header for quick keyword searches
- Filter search results by category
- Return search results with the closest match at the top and descending

Order Management Service

- Creates original order ID at checkout
- Displays Order History
- Re-order previous orders

Delivery Service

- Validation for client delivery address
- Confirms the address is within the location radius.
- Alerts if address is too far and stops from proceeding further

Checkout Service

- Gathers order billing and delivery information or pickup time

- Sends out payment to external
- Is able to handle canceled/failed transactions
- checkout completion confirmation

Ordering Cart

- Creation of user cart
- Stores selected items, customizations, and quantity.
- Calculations for order total, fees, and tips.

Data

- Stores data for the menu, ingredients, and allergens in dishes
- Stores uses and delivery addresses
- Stores users carts
- Stores users previous order details

External

- External payment processing gateway
- Location system such as a geocoding api
- Email notification and contact services

2. Written requirements specifications

A. All functional requirements (FRs)

Menu & Display Services

FR : Menu Loading

The system shall receive and display all available menu items from the database upon user

access.

FR : Category Grouping

The system shall group menu items by category.

FR : Item Details

Then a user selects a menu item, the System shall display the item's name. Description, price, ingredient list, allergen data, and availability status.

FR : Out of Stock Restriction

If a menu item is marked as unavailable, the system shall prevent the item from being added to the cart.

Search & Filtering

FR : Keyword Search

The system shall allow users to search for menu items using a keyword search bar located in the header.

FR : Ranked Results

The system shall return search results ordered by relevance to the entered keyword.

FR :Allergen Exclusion

When a user selects one or more allergens, the system shall remove menu items containing those allergens from the displayed results.

FR : Allergen Warning

The system shall flag items that " may contain" a selected allergen.

FR : Dietary Filter

The system shall allow filtering of items based on dietary restrictions like vegetarian and gluten free.

Item Customization

FR : Customization Options

The system shall allow users to select predefined customization options for applicable menu items.

FR : Dynamic Price Update

The system shall update the item price immediately when customization options are selected.

FR : Cart Attachment

The system shall attach selected customization options to the corresponding cart item.

Cart Management

FR : Add to Cart

The system shall allow users to add menu items to a cart.

FR : Modify Cart

The system shall allow users to update item quantity or remove items from the cart.

FR : Order Calculations

The system shall calculate subtotal, taxes, delivery fees, service fees, and optional tip amounts.

FR : Cart Integrity

The system shall not allow checkout if the cart is empty.

Checkout & Delivery

FR : Information Collection

The system shall collect delivery or pickup information during checkout.

FR : Field Validation

The system shall validate required fields before allowing payment processing.

FR : Address Validation

The system shall validate the delivery address using an external geocoding service.

FR : Delivery Radius Enforcement

If the address is outside the service radius, the system shall block checkout and notify the user.

Payment & Order Processing

FR : Payment Transmission

The system shall transmit payment information securely to an external payment processor.

FR : Payment Failure Handling

If payment fails, the system shall not create an order record.

FR : Order ID Generation

Upon successful payment, the system shall generate a unique order ID.

FR : Order Storage

The system shall store order details in the database after successful checkout.

FR : Confirmation Display

The system shall display an order confirmation page after successful checkout.

FR : Email Notification

The system shall send an email confirmation after successful checkout.

B. All non-functional requirements (NFRs)

Product Requirements

Performance

NFR : Search Response Time

Search results shall be returned within 1 second for a menu containing up to 500 items.

NFR : Checkout Processing Time

Checkout confirmation shall be displayed within 3 seconds of receiving a successful payment response.

Reliability

NFR : Order Integrity

The system shall prevent duplicate order creation during network interruptions.

NFR : Availability

The system shall maintain 99% availability during operating hours.

Security

NFR : Encryption

All client-server communication shall use HTTPS encryption.

NFR : Payment Data Protection

The system shall not store raw credit card information.

NFR : Input Validation

The system shall validate and sanitize all user inputs.

Usability

NFR : Responsive Design

The system shall support mobile, tablet, and desktop screen sizes.

NFR : Checkout Simplicity

Users shall be able to complete checkout in no more than five steps.

NFR : Accessibility Compliance

The system shall comply with WCAG 2.1 Level AA accessibility standards.

Organizational Requirements

NFR : Development Standards

The system shall be developed using RESTful API architecture.

NFR : Documentation

All deliverables shall conform to ITIS 3300 documentation standards.

NFR : Maintainability

The system architecture shall be modular to allow future feature enhancements.

External Requirements

Interoperability

NFR : Payment Integration

The system shall integrate with an external payment processing API.

NFR : Geocoding Integration

The system shall integrate with a geocoding API for address validation.

Privacy & Legislative

NFR : Data Privacy

The system shall allow users to review and correct their stored contact information.

NFR : Data Protection Compliance

The system shall comply with applicable data protection regulations.

C. Interfaces

User Interface

- Responsive web Interface
- Header search bar
- Allergen and dietary filter controls

- Customization modal
- Cart Interface
- Checkout form with validation
- Confirmation Screen

Software Interfaces

- GET /menu
- GET /search
- POST /cart
- POST /checkout
- GET /orders

External Interfaces

- External payment processing API
- Geocoding API
- Email Service API

3. The overall plan for implementing the project through three development phases. For each development phase, the corresponding requirements should be identified (need to prioritize functionalities based on their criticality).

3. Development Phase Plan

The project will be implemented using an incremental development approach across three phases. Each phase prioritizes core system functionality before expanding to advanced features and refinements.

Phase 1 – Core Ordering Functionality

Objective: Make sure the system can display the menu, allow orders, and process payments.

Included Requirements:
FR: Menu Loading
FR: Category Grouping
FR: Item Details
FR: Add to Cart
FR: Modify Cart
FR: Order Calculations
FR: Cart Integrity
FR: Information Collection
FR: Field Validation
FR: Payment Transmission
FR: Order ID Generation
FR: Order Storage
FR: Confirmation Display

Criticality:
These requirements are necessary for the system to work. Without them, users would not be able to place orders.

This phase allows the system to:

- ➔ Display menu items
- ➔ Allow ordering
- ➔ Process payments
- ➔ Store orders

Phase 2 – Enhanced user experience

Objective: Enhance the core system by improving search functionality

Included Requirements:

FR: Keyword Search
FR: Allergen Exclusion
FR: Customization Options
FR: Address Validation
FR: Email Notification

Criticality:

These features improve usability and delivery accuracy

This phase allows the system to:

- ➔ Improve item discovery through keyword search
- ➔ Filter menu items based on allergen selections
- ➔ Provide dish customization options
- ➔ Validate delivery addresses before checkout

Phase 3 - Refinements, Edge Cases, and Advanced Filtering

Objective: Finalize the system by handling operational edge cases, advanced filtering, and other extra features.

Included Requirements:

FR: Out of Stock Restriction
FR: Ranked Results
FR: Allergen Warning
FR: Dietary Filter
FR: Dynamic Price Update
FR: Cart Attachment
FR: Delivery Radius Enforcement
FR: Payment Failure Handling

Criticality:

These requirements are prioritized last because they build upon the foundational features established in Phases 1 and 2. While core ordering and basic search must function first, Phase 3 features are critical for real-world business operations. Handling payment failures, enforcing delivery radii, and restricting out-of-stock items protect the business from revenue loss and

operational errors. Meanwhile, dynamic price updates and advanced dietary filters provide the necessary polish for a fully finalized user experience.

This phase allows the system to:

- ➔ Prevent orders for out-of-stock items
- ➔ Handle payment failures securely
- ➔ Enforce delivery boundaries
- ➔ Update prices dynamically
- ➔ Provide advanced dietary and allergen warnings